SpotCast

Synopsis for the app project

SWMAD-01 Mobile Application Development

Group members and study numbers: [FILL IN: name, study number, one line per member]

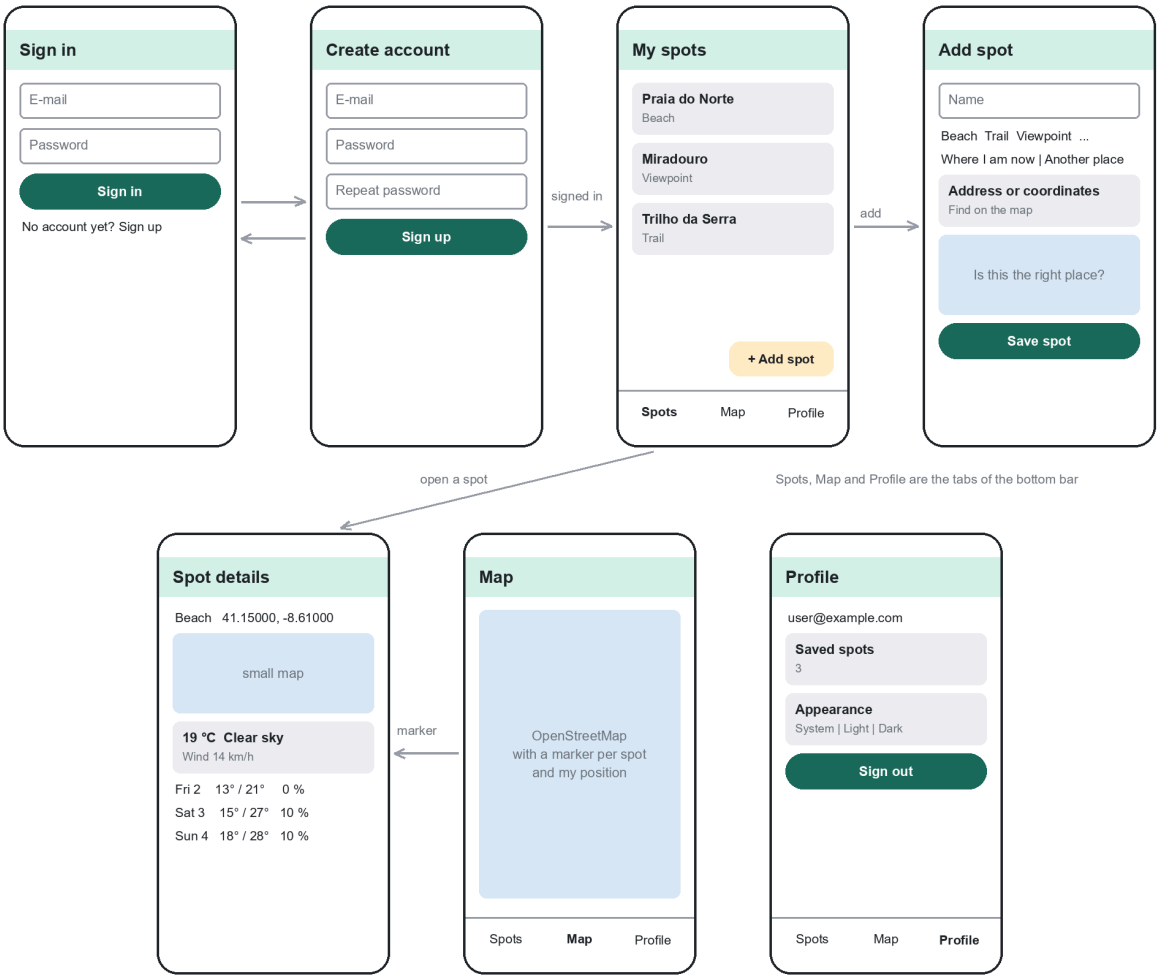
App vision

People who spend time outdoors keep a mental list of places: a beach that works in a westerly wind, a trail head with parking, a viewpoint worth the climb. Before going they check the weather for each place by typing its name into a weather app, which often gives the forecast for the nearest town and not for the place itself.

SpotCast lets the user save a spot at the position where they are standing, with a name, a category and a note. The spots are stored in the user's account and shown in a list and on a map. Opening a spot shows the current weather and a seven-day forecast for its exact coordinates.

Screens

The app has five screens besides sign in and sign up. The layout below shows all seven.



- **Spots**: the list of the user's spots and a button to add one.
- **Add spot**: name, category, note and the position, either from GPS or from an address or coordinates that the user confirms on a map.
- **Spot details**: the spot, a small map, the weather now and the next seven days.
- **Map**: all spots as markers, and the user's own position.
- **Profile**: the account, the number of spots, the choice between a light, a dark and the system theme, and sign out.

User stories

- As a hiker, I want to save the place where I am standing with one button, so that I do not have to type coordinates or search for an address.
- As a surfer, I want to see the forecast for the exact position of my spot, so that I can decide whether the trip is worth it.
- As a user with several spots, I want to see them on a map, so that I can pick the ones that are close to each other for the same day.
- As someone planning a trip, I want to save a spot by its address or coordinates and check it on a map, so that I can follow its forecast before I have been there.
- As a user with a phone and a tablet, I want my spots to follow my account, so that a spot saved on one device shows up on the other without me doing anything.

Technologies

- Kotlin, coroutines and Jetpack Compose with Compose Navigation.
- Firebase Authentication with e-mail and password (mandatory).
- Firestore database for the spots of each user (mandatory).
- Ktor HTTP client calling the Open-Meteo forecast API, which is not a Firebase service and is called without an SDK (mandatory).
- GPS through the fused location provider, and maps from OpenStreetMap with the osmdroid library (the additional technology).
- Firestore real-time updates for the spots list.
- Address search with Nominatim (OpenStreetMap), also through the Ktor client.